

Object Oriented Programming In Java

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Why to learn Object Oriented Programming in Java

?

5 WHY Technique to get the answer

Course Content

Prerequisites:

Programming skill is not mandatory but it is recommended to have basic knowledge of Introduction to Programming. In addition, basic knowledge of command line shells is preferred but not necessary. If you already know C or any other programming language, Java should be easy to pick up.

Course Learning Outcomes

- CLO1 : Understand the basic concept of Object Oriented programming in Java. In addition, students will be able to write simple programs
- CLO2 : Learn about variables and several data types. In addition, they will also learn about encapsulation through programming examples

Course Content

- CLO3 : Learn the concepts of constructors, abstraction, inheritance and polymorphism
- CLO4 : A quick recap of if-else, loops in programming and learning about their implementation in Java
- CLO5 : Recap of the concept of recursion and its implementation in Java
- CLO6 : Learn String handling in Java and Java APIs
- CLO7 : Learn the advance classes for data storage, manipulation and retrieval and java basic data structures
- CLO8 : to consider user inputs and process them. In addition, they will also get accustomed to handling exceptions in programs.

Course Content

- CLO9 : Learn to work with files: consider input from files, process them and store outputs to different files. They will also be able to work with directories that contain multiple files.
- CLO10 : Learn to use Java to connect to a database system. They will be able to use Java API to connect and execute the query with the database.
- CLO11 : Learn multithreading concepts in java

Study Material

Textbooks:

Java: A Beginner's Guide, 9th Edition by Herbert Schildt

Head First Java, 3rd Edition by Kathy Sierra

Additional Study Material

<https://www.learnjavaonline.org/>

<https://docs.oracle.com/javase/tutorial/>

History of Java

- Java was conceived by James Gosling, Patrick Naughton, Chris Warth, Ed Frank, and Mike Sheridan at Sun Microsystems in 1991. This language was initially called “Oak” but was renamed “Java” in 1995.
- The primary motivation was the need for a platform-independent language that could be used to create software to be embedded in various consumer electronic devices, such as toasters, microwave ovens, and remote controls. About the time that the details of Java were being worked out, a second, and ultimately more important, factor emerged that would play a crucial role in the future of Java. This second force was, of course, the World Wide Web. Had the Web not taken shape at about the same time that Java was being implemented, Java might have remained a useful but obscure language for programming consumer electronics.
- Java is directly related to both C and C++. Java inherits its syntax from C. Its object model is adapted from C++. Java’s relationship with C and C++ is important for several reasons.

Class and Objects

What is Class ?

A class is the **BLUEPRINT** of an object. A class defines the shared characteristics like -

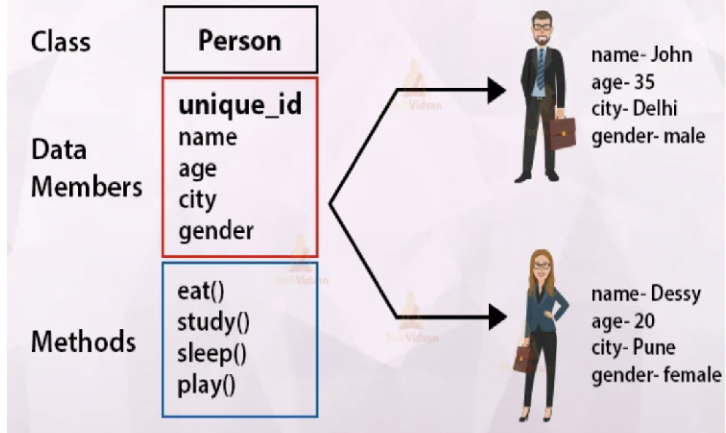
- The set of attributes/properties - instance variables
- The set of behaviour or actions - methods
- How to construct an object - constructors

What is Object ?.

Objects are the “instances” of a Class. Any entity that has state and behaviour is known as an object.

Class and Objects

Java Class & Objects



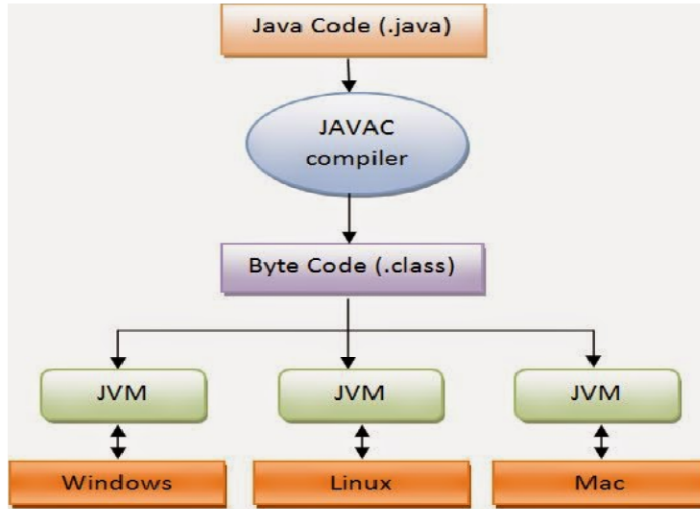
What is OOPS and Its Advantages ?

Object-oriented programming System (OOPS) is based on the concept of objects. Whenever we develop some programs using Class and Objects than it is called Object Oriented Programming System. Java is an object-oriented programming language.

Advantages Over Procedural Programming:

- OOP is faster and easier to execute.
- OOP provides a clear structure for the programs.
- OOP keeps code easier to maintain, modify and debug.
- OOP make it possible to create full reusable applications with less code and shorter development time.

Portability : Write once, use anywhere



Practical: JDK and Eclipse installation

- https://www3.ntu.edu.sg/home/ehchua/programming/howto/EclipseJava_HowTo.html
- Write Hello World program
- Write Student Class(Run through command line too)